

Installation Guide

Course: DLBDSPBDM01 – Build a Data Mart in SQL; **Programme:** B.Sc. Computer Science; **Author:** Zubair Ahmed Khan; **Matriculation Number:** 92127983; **Tutor:** Prof. Dr. Anna Androvitsanea; **Submission Date:** 2026-07-29

1. Purpose

This document describes how to install MySQL Server and MySQL Workbench, and how to deploy the **AirbnbDB** database on a local machine. Following the steps below will create the database, generate all twenty-one tables, establish the primary-key and foreign-key relationships, and load the dummy data required to run the three test cases.

2. Prerequisites

The following software must be installed before running the SQL script.

Software	Minimum Version	Purpose
MySQL Server	8.0	Hosts the AirbnbDB schema and stores all data.
MySQL Workbench	8.0	Graphical client for opening and executing the SQL script.
Operating System	Windows 10/11, macOS 11+, or Ubuntu 20.04+	Runs the MySQL software.
Disk Space	Approx. 500 MB	Stores the MySQL binaries, the schema, and the dummy data.
RAM	4 GB minimum	Sufficient for the small data volume of this coursework project.

A standard broadband connection is needed only if MySQL is downloaded for the first time. The AirbnbDB script itself is fully self-contained and does not require an internet connection at execution time.

3. Installing MySQL Server and MySQL Workbench

3.1 Windows

1. Visit the official MySQL download page at <https://dev.mysql.com/downloads/installer/>.
2. Download the **MySQL Installer for Windows** (the larger "Full" bundle is recommended).
3. Launch the installer and choose **Developer Default** as the setup type. This option installs MySQL Server, MySQL Workbench, MySQL Shell, and the sample documentation together.
4. When prompted, set a **root password** and record it securely. A username/password combination is required later in Step 4.

5. Complete the installation and leave the "Start MySQL Workbench" box checked.

3.2 macOS

1. Download the **MySQL Community Server DMG** from <https://dev.mysql.com/downloads/mysql/>.
2. Install the package and, during the configuration step, choose "Use Legacy Password Encryption" and set a root password.
3. Download **MySQL Workbench for macOS** from the same download portal and drag the application to the Applications folder.

3.3 Linux (Ubuntu / Debian)

1. Open a terminal and run the following commands in sequence:

```
bash sudo apt update sudo apt install mysql-server sudo mysql_secure_installation
```

```
bash sudo snap install mysql-workbench-community
```

4. Deploying the AirbnbDB Schema

Step 1 — Open MySQL Workbench

Launch MySQL Workbench. In the **MySQL Connections** panel, click the local instance created during installation (commonly labelled `Local instance 3306`). Enter the root password when prompted. A successful connection opens the SQL editor tab.

Step 2 — Open the SQL Script

In the Workbench menu, choose **File** → **Open SQL Script...** and navigate to the file `airbnb.sql` delivered alongside this document. The full script appears in the editor. No code changes are required.

Step 3 — Execute the Script

Click the **Execute** button (the lightning-bolt icon) or press `ctrl+shift+Enter` (`Cmd+Shift+Enter` on macOS). MySQL runs every statement in order, beginning with `SHOW DATABASES`; and continuing through the `CREATE`, `ALTER`, and `INSERT` statements.

Step 4 — Confirm Database Creation

In a fresh query tab, run the following statement to confirm the database is present:

```
<code class="language-sql">SHOW DATABASES;
```

Expected output (one of the rows):

```
| AirbnbDB |
```

Step 5 — Confirm Table Creation

Run the following statement:

```
<code class="language-sql">USE AirbnbDB;
```

```
SHOW TABLES;
```

Expected output: A list of twenty-one tables, including Accommodation, Amenity, AmenityListing, Booking, Complaint, Employee, Guest, Host, HouseRules, Income, Language, Listing, Location, Media, Notification, Payment, Place, Reservation, Review, User, and UserLanguage.

Step 6 — Confirm Dummy Data

Run any of the following verification statements to confirm that at least twenty records exist in each table:

```
<code class="language-sql">SELECT COUNT(*) FROM User;
```

```
SELECT COUNT(*) FROM Guest;
```

```
SELECT COUNT(*) FROM Host;
```

```
SELECT COUNT(*) FROM Accommodation;
```

```
SELECT COUNT(*) FROM Booking;
```

```
SELECT COUNT(*) FROM Payment;
```

```
SELECT COUNT(*) FROM Reservation;
```

```
SELECT COUNT(*) FROM Review;
```

```
SELECT COUNT(*) FROM Complaint;
```

```
SELECT COUNT(*) FROM Notification;
```

```
SELECT COUNT(*) FROM Location;
```

```
SELECT COUNT(*) FROM HouseRules;
```

```
SELECT COUNT(*) FROM Place;
```

```
SELECT COUNT(*) FROM Amenity;
```

```
SELECT COUNT(*) FROM AmenityListing;
```

```
SELECT COUNT(*) FROM Media;
```

```
SELECT COUNT(*) FROM Listing;
```

```
SELECT COUNT(*) FROM Employee;
```

```
SELECT COUNT(*) FROM Language;
```

```
SELECT COUNT(*) FROM UserLanguage;
```

```
SELECT COUNT(*) FROM Income;
```

Expected output: every query returns a count of at least 20.

Step 7 — Run the Three Test Cases

Copy and execute the three test cases documented in the **Test Case Document** of the submission:

- **Test Case 1** — Retrieval of complete booking information.
- **Test Case 2** — Calculation of host accommodation statistics and income.
- **Test Case 3** — Retrieval of complete accommodation information.

Expected output: Each test case returns a populated result set that joins multiple tables. The presence of rows confirms that the foreign-key relationships are intact and that the dummy data is consistent.

5. Database Metadata at a Glance

After a successful installation, the database contains the following:

Item	Value
Database name	AirbnbDB
Number of tables	21
Minimum records per table	20
Foreign-key relationships	Active
Test cases executable	3

6. Basic Troubleshooting

Problem	Likely Cause	Resolution
"Access denied for user 'root'@'localhost'"	Incorrect root password.	Re-enter the password or reset it through <code>mysql_secure_installation</code> .
"Unknown database 'AirbnbDB'"	The script was not executed in full.	Re-open <code>airbnb.sql</code> and click Execute from the top of the file.
"Cannot add foreign key constraint"	Tables were created out of order.	Drop the database with <code>DROP DATABASE AirbnbDB;</code> and re-execute the script from the beginning.
<code>SELECT COUNT(*)</code> returns 0	The <code>INSERT</code> statements did not run.	Scroll up in the Action Output panel and check for errors; re-execute the <code>INSERT</code> block.
Workbench does not start on Linux	Missing libraries.	Install the dependencies with <code>sudo apt install libmysqlclient21 libgtk-3-0</code> .
"Data too long for column"	A field value exceeds its declared length.	Re-open the script, locate the failing row, and either shorten the value or widen the column type.
Script runs but <code>SHOW TABLES</code> is empty	A previous attempt created a partial schema.	Run <code>DROP DATABASE IF EXISTS AirbnbDB;</code> and re-execute the script from the top.

7. Uninstallation

To remove the database, run the following statement in MySQL Workbench:

```
<code class="language-sql">DROP DATABASE AirbnbDB;
```

To remove MySQL Server and MySQL Workbench themselves, use the standard uninstaller of the host operating system. On Windows, the MySQL Installer provides a "Remove" option; on macOS, the DMG includes an uninstall package; on Linux, use `sudo apt remove mysql-server mysql-workbench-community`.

8. Support

For questions specific to this project, refer to the abstract, the SQL documentation, the test case document, and the data dictionary. For questions about MySQL itself, the official MySQL Reference Manual at <https://dev.mysql.com/doc/refman/8.0/en/> is the authoritative resource.